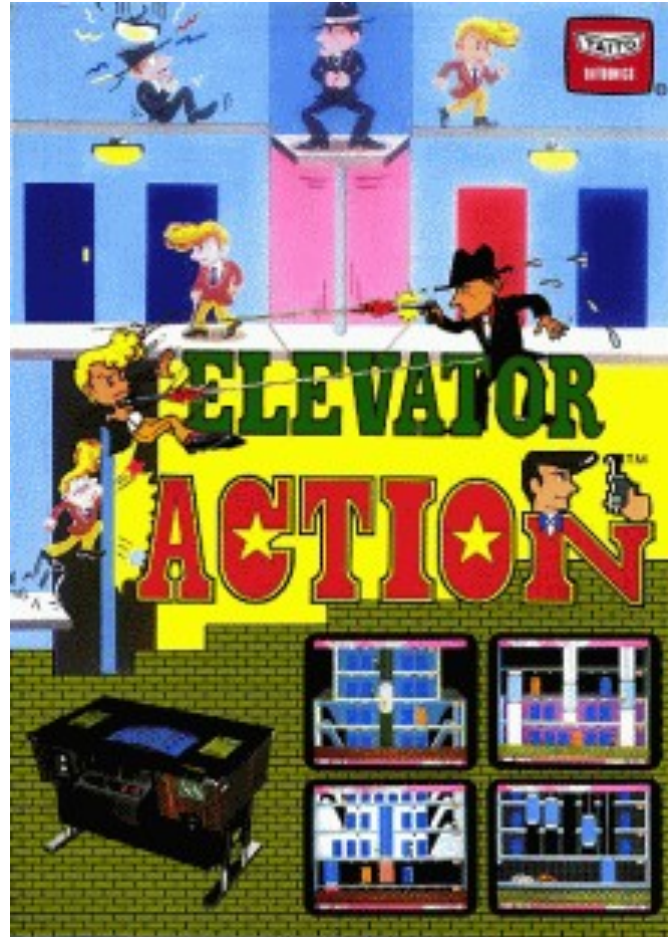


Phreaking Elevators

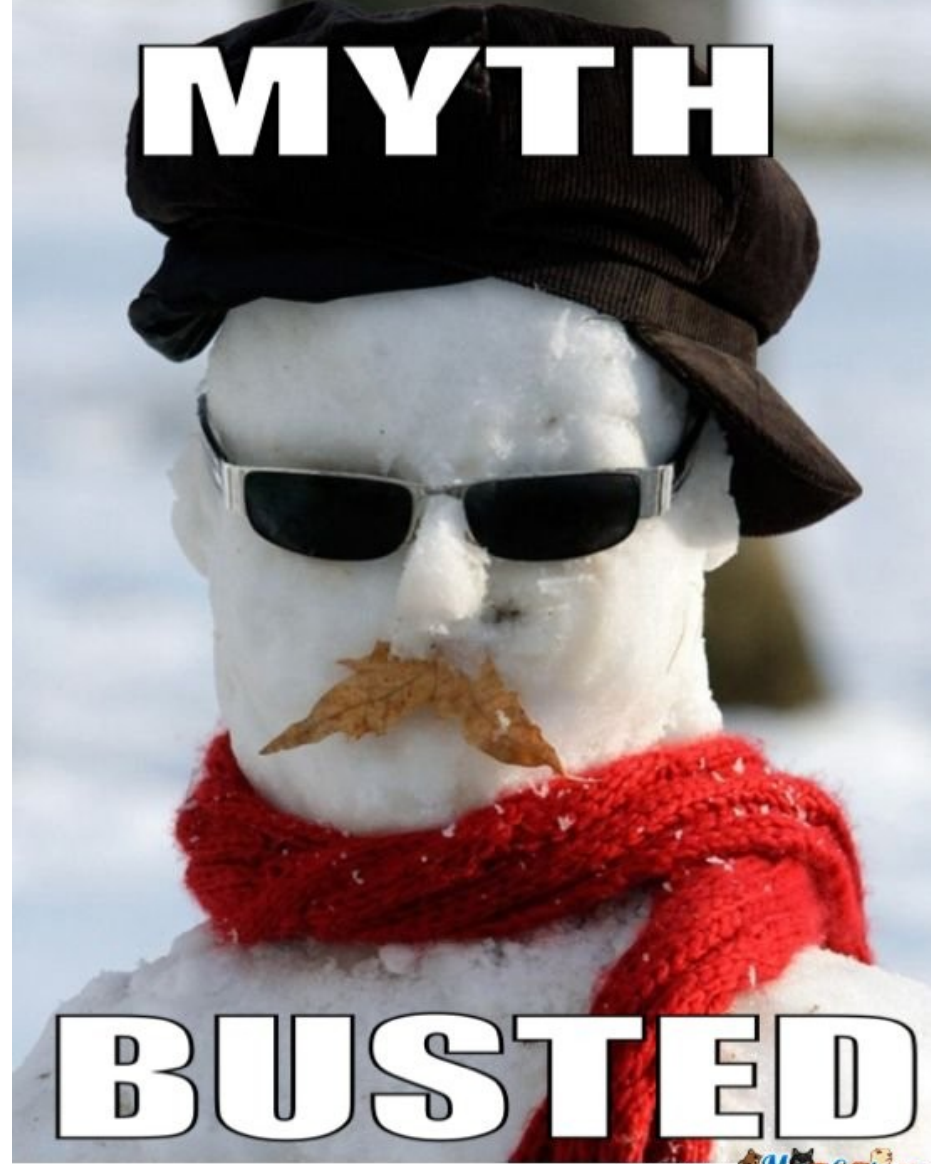


Who am I?

- Car hacking things
- High voltage projects
- CTFs (lifetime badge winner for Cyphercon)
- Infosec is a hobby (I might be looking for a job)
- I am new to the infosec world (3 years)



**You can not
control an
elevator though
the phone.**





Want to know more on elevators?

- Deviant Ollam & Howard Payne - Elevator Hacking - From the Pit to the Penthouse
(Watch the talk they did at HOPE)

(We will have more from Deviant later on that guy is awesome.)

Topics we are going to be covering:

- Elevator phone systems basics
- History (not a lot of this since its 2019 I want about things that you can do TODAY)
- Information gathering
- Tools I like to use
- Elevator phone systems more advanced (you better be ready)
- Thinking out side the elevator
- Conclusion

Legal disclaimer

- I am not a lawyer please don't hack things you don't own.
- None of the recordings involve a live human talking on the other end.
- Don't interfere with emergency communications these systems are in place for a good reason.

Please don't do this in the wild...



Elevator phone systems basics



04

NO PERSONS OTHER
THAN THE OPERATOR
THE FREIGHT HANDL
ARE PERMITTED TO
ON THIS ELEVATOR

CLASS C-1 LOADING

THIS ELEVATOR DESIG
TO TRANSPORT LOA
INDUSTRIAL TRUCK
MAXIMUM COMBIN
WEIGHT OF INDUSTR
TRUCK AND LOAD
NOT TO EXCEED 12,000

CALL CANCEL

CLASS C-1 LOADING. THIS
ELEVATOR DESIGNED TO
TRANSPORT LOADED INDUSTRIAL
TRUCK. MAXIMUM COMBINED
WEIGHT OF INDUSTRIAL TRUCK
AND LOAD NOT TO
EXCEED 12,000 LB.

3 4
1 2
K
A

CALL

CAPACITY
12,000 LBS.

KONE

ELEVATOR NO. 2

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100

YOU MAY SPEAK WHEN LIGHTS ON

CALL

PUSH
TO CALL

MOMENTARY LIGHT INDICATES
ALARM RECEIVED

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100

CAPACITY 2500 LBS.

EMERGENCY PHONE

PUSH TO TALK

RED LIGHT - EMERGENCY MESSAGE PLAYING
BLINKING LIGHT - EMERGENCY MESSAGE RECEIVED

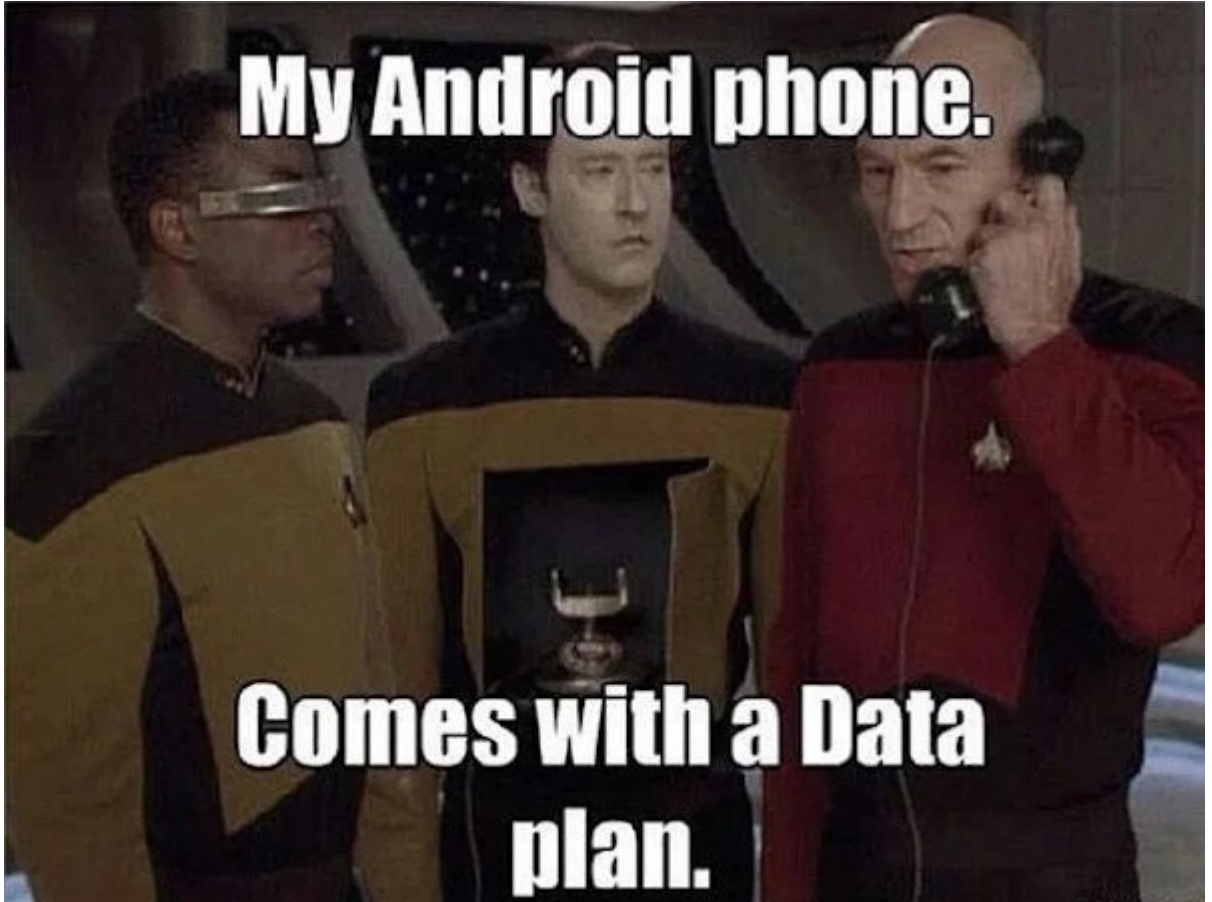
BATH MICROPHONE
BATH MICROPHONE





To the outside world

- POTS
- VOIP
- Cell



Elevator phone systems basics

- POTS (plain old telephone service) this is in most elevators it's how a phone will dial out.
- The device will dial when the hand set is picked up or when the button is pushed.
- ADA/ASME A17 is the reason we have changes in our elevator phones. Also there are building codes that need to be followed.

History time!

- When elevators phones started in 1968.
- We still find rotary phones!
- Ring down system.
- It was required for all elevators in 1976.

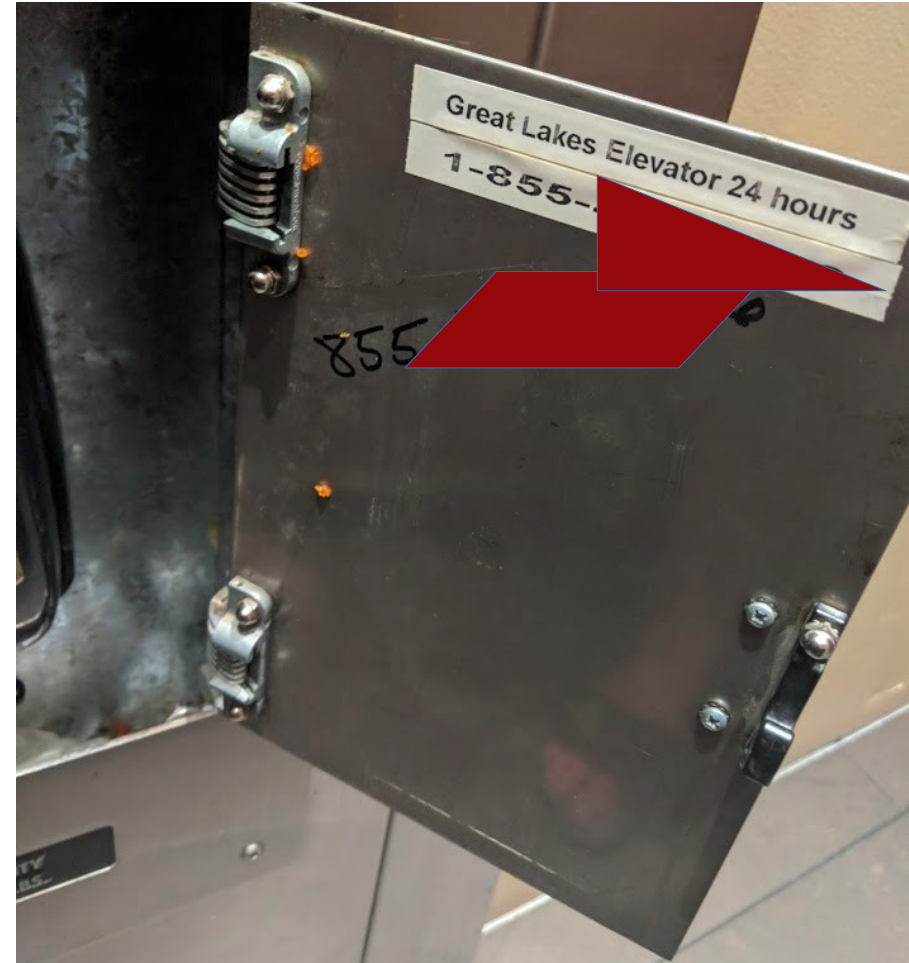




Information gathering

OSINT

- Who the phone dials to
- Building information
- Phone system information
- Phone number information
(OpenCNAM)
- Google





Rahm Emanuel
Mayor

City of Chicago
Department of Buildings

No.

CONDITIONAL CERTIFICATE of INSPECTION

DATE OF INSPECTION: 6-15-18

DATE OF EXPIRATION: 8-15-18

THIS CERTIFIES THAT I HAVE THIS DAY INSPECTED THE

PASSENGER ELEVATOR

at

☐

Certificate pending. Found to be in safe operation.

☒

Due to lack of compliance with ALL code requirements this certificate is considered conditional. Further inspection will be required to ensure full compliance with the code.

☐

PERMIT REQUIRED

REPAIR PERMIT#:

☐

NO PERMIT REQUIRED

This inspection is performed to comply with code section: 13-20-100-Inspection Required.

This certificate is posted in compliance with code section: 13-20-110-Certificate of Compliance-Posting.

Yasha 533
ELEVATOR INSPECTOR

Judith Frydland
Commissioner
Judith Frydland

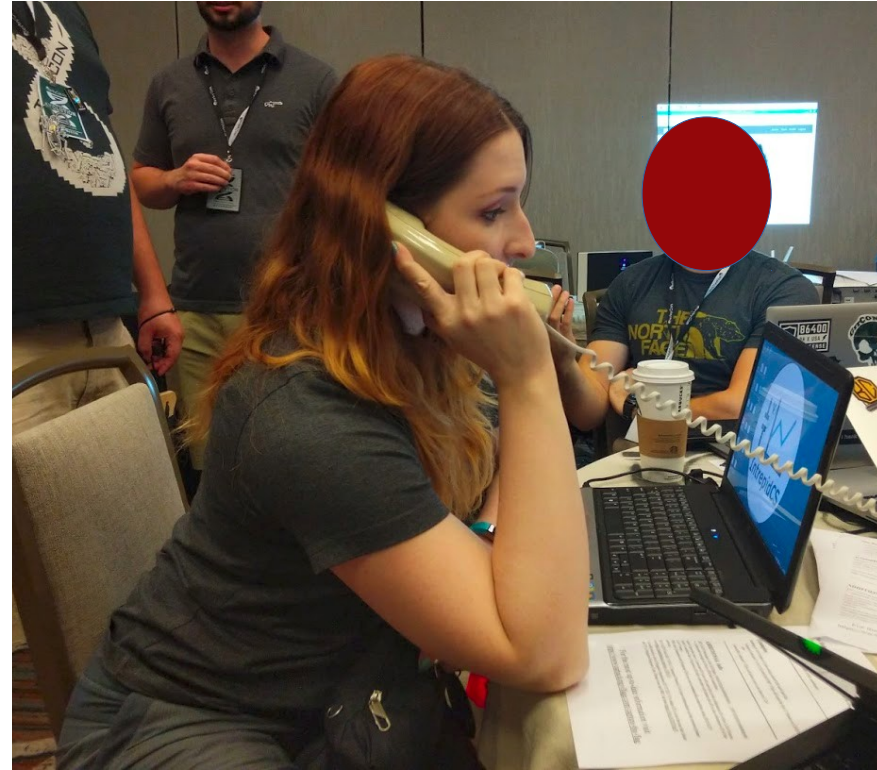
THIS CERTIFICATE MUST BE DISPLAYED IN THE ELEVATOR CAB



Information gathering

Social engineering

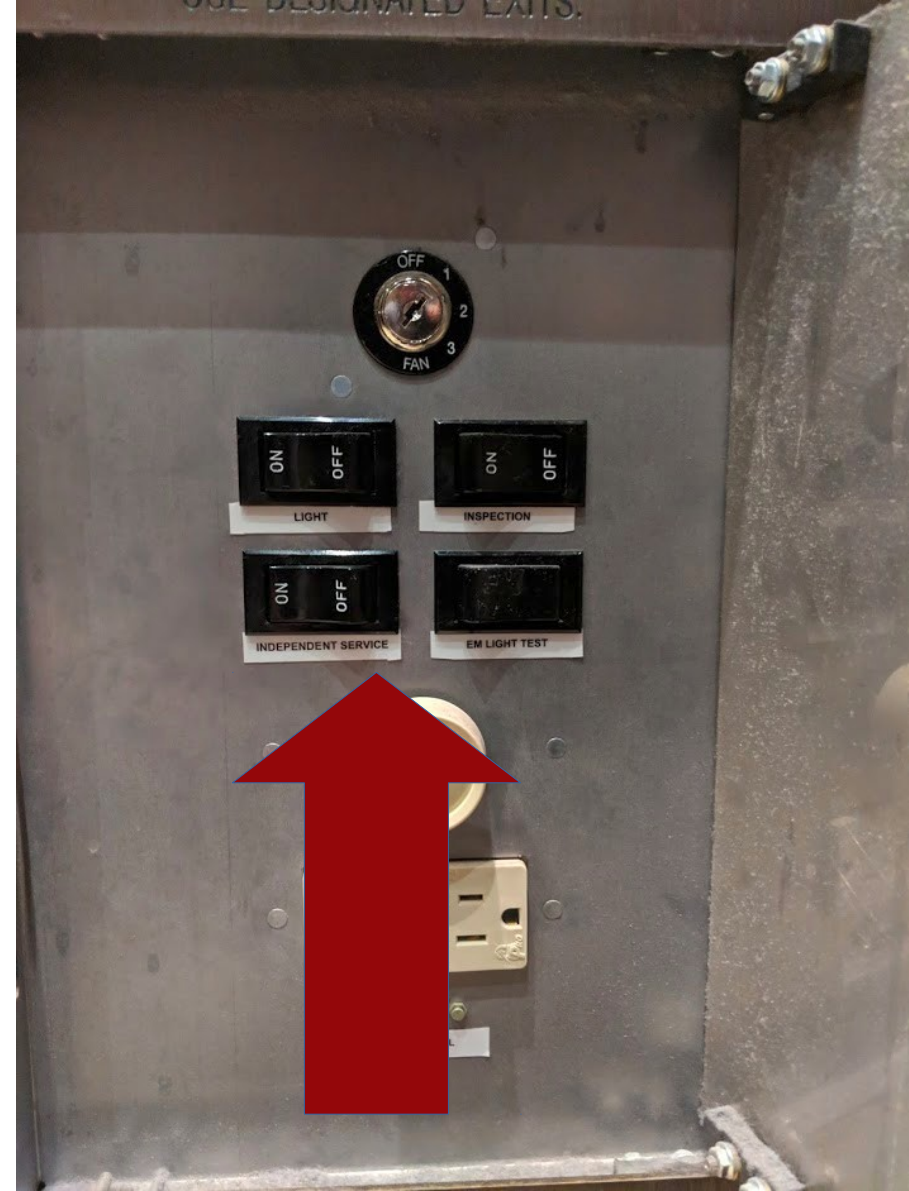
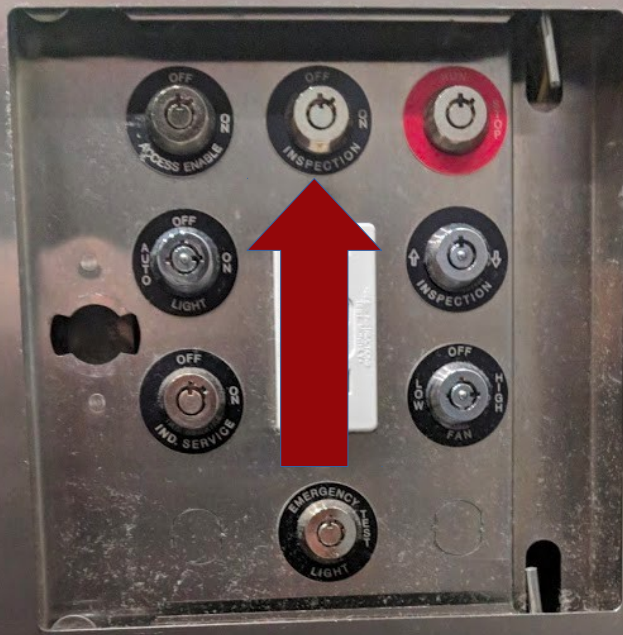
- Pretend to be an elevator tech
- Remotely gathering target information by SEing the business and call centers



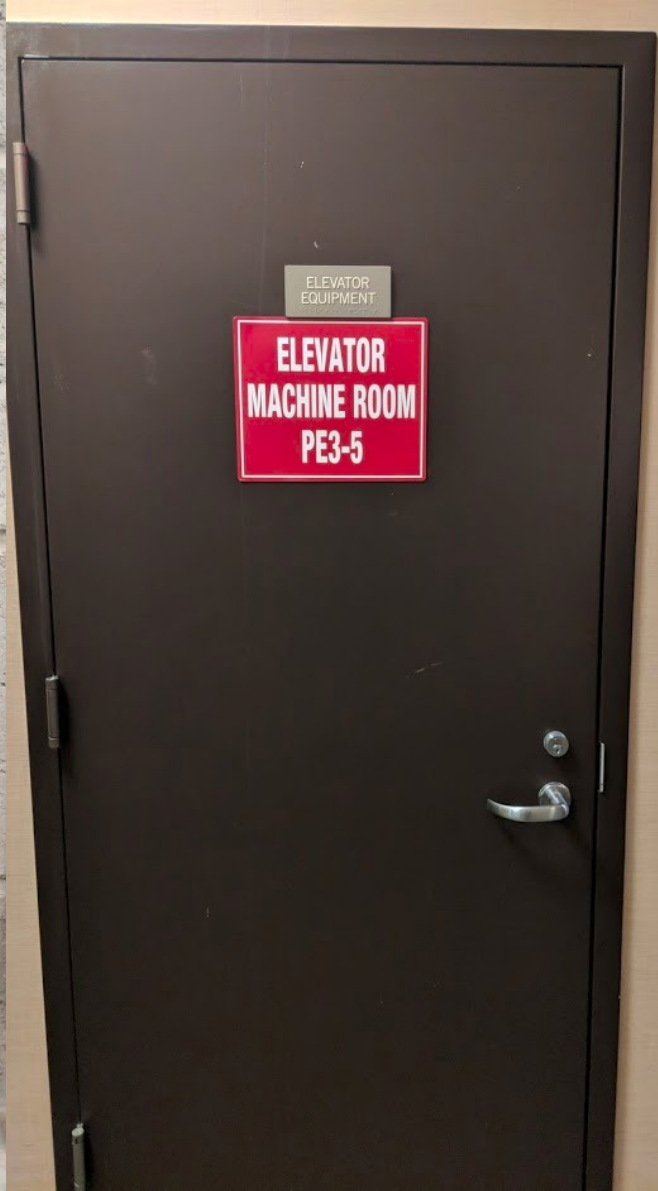
Information gathering

Physical

- How to control an elevator with independent service mode
- Where to go and what to look for







TOOLS

- Cell Phone
- Multi tool
- Land line phone
- Linesmen handset
- 9volt battery
(some times a AA)



Few good points

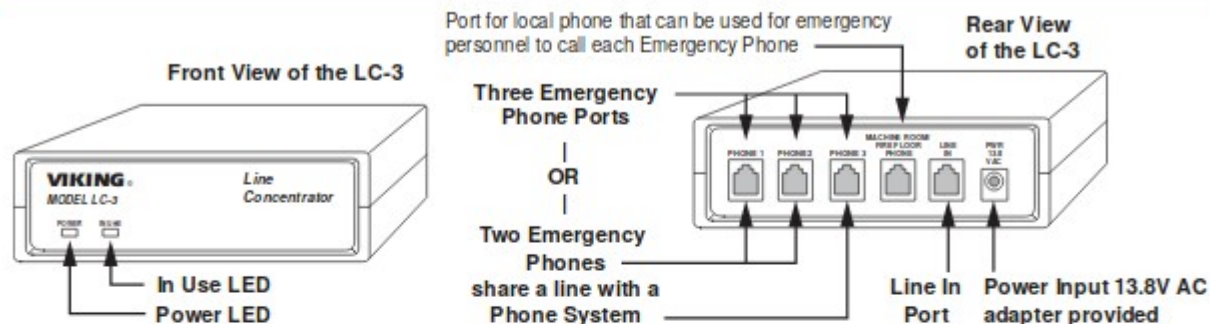
- Site ID 2
- Hang up *# or # or *0 or 0
or (press the button again)
- PBX and line concentrators



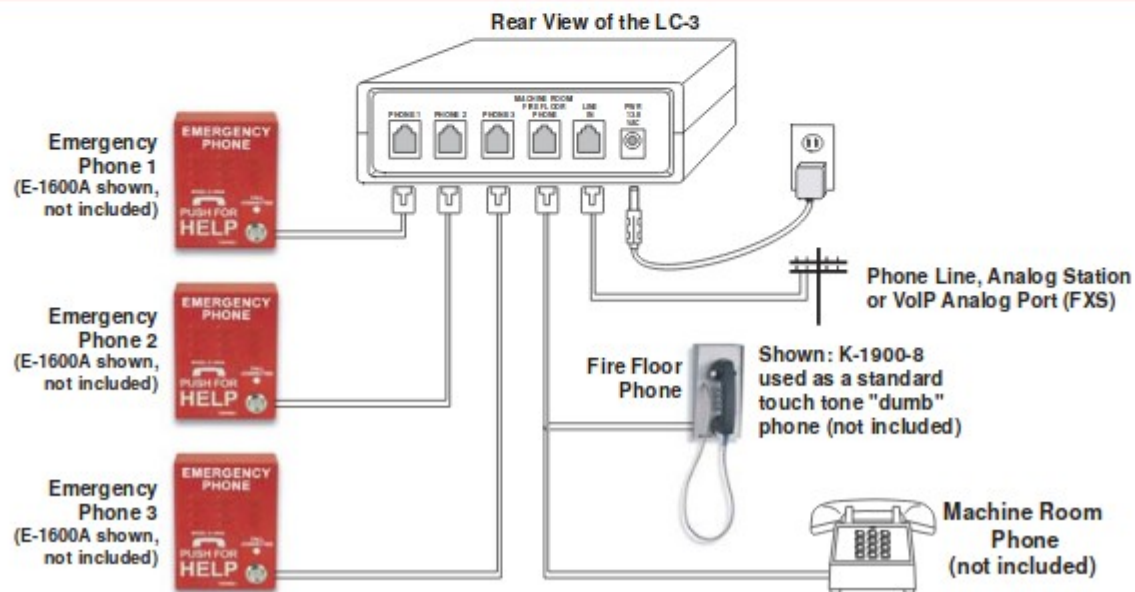
PBX

- Private branch exchange
- Elevators can be on their own PBX
- Older PBXs you can find from a low voltage (24Volts)

Features Overview



Installation



Line Concentrator

- Outbound one call at a time the other get a busy signal
- Inbound Call the LC pick up with a double beep
- With in 4 seconds enter “1”, “2” or “3” before it drops the line.
- When the power fails its a party!

Intercom

- Operated from outside the elevator
- Located in hall way, hostway or machine room
- Video time!



Video!



Fire Fighters Phone

- The system looks for opens, shorts, or ground faults.
- Class A and Class B wiring
- Sounds an alert at fire command center until answered.



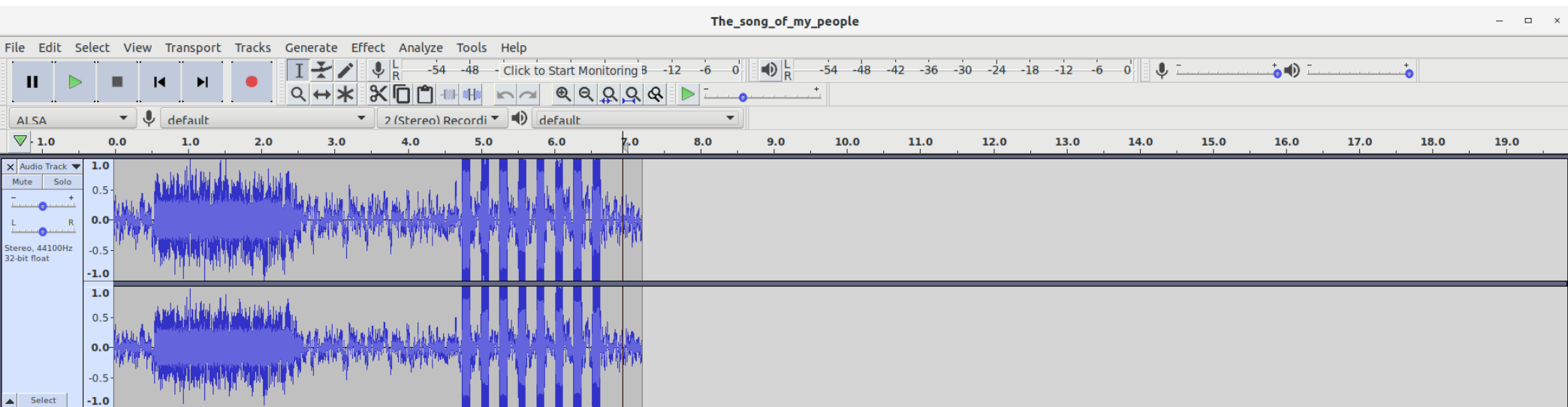


Line detection

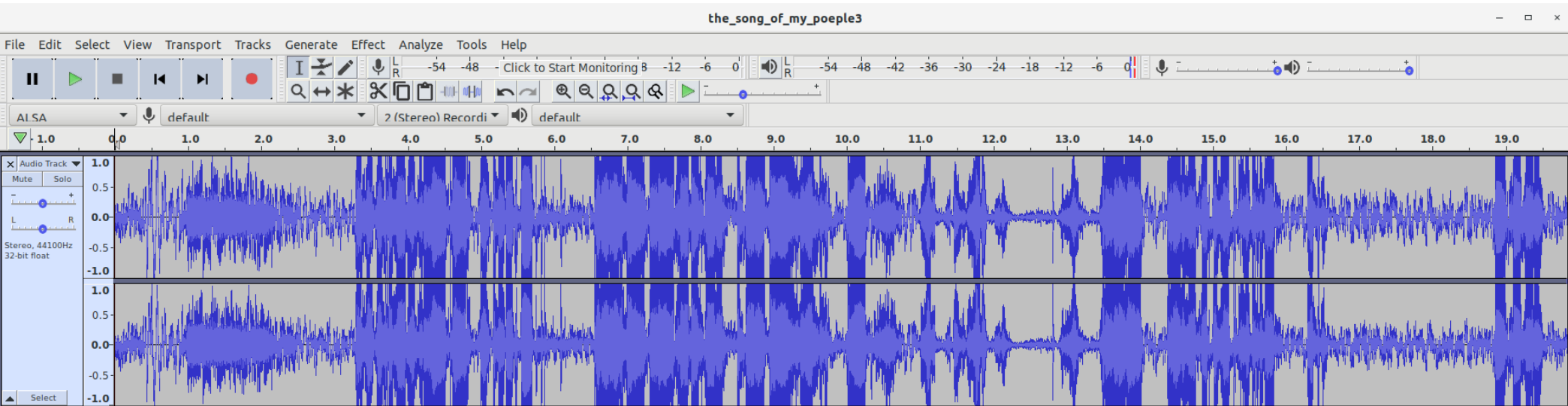
- Checking on the phone line
- The key can reset the device for 24 hours
- The key can control volume
- **RTFM**



The song of my people....

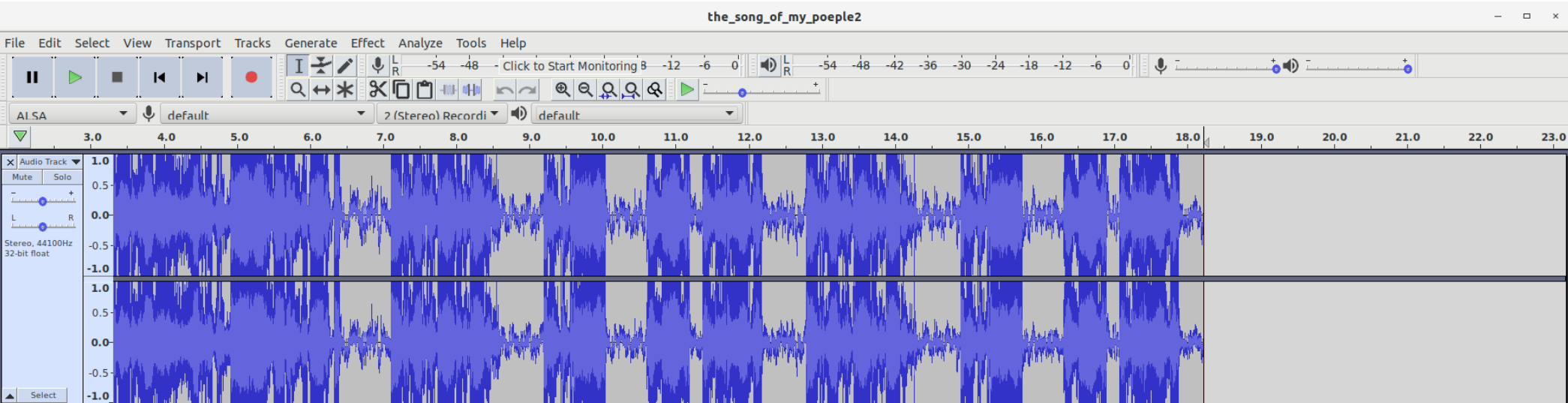


The song of my people....



This an emergency telephone located at. This is elevator emergency inside car one at _____ main st please wait for two way communication thank you. Main menu press 1 to talk 2 to program or *0 to disconnect.

The song of my people...



This the automated emergency telephone located at
Press 1 to talk to the party 2 to program 0 to disconnect
Press 1 to talk to the party 2 to program 0 to disconnect

Four ways to program

- Key pad
- Switches
- Remote
- Programming cable

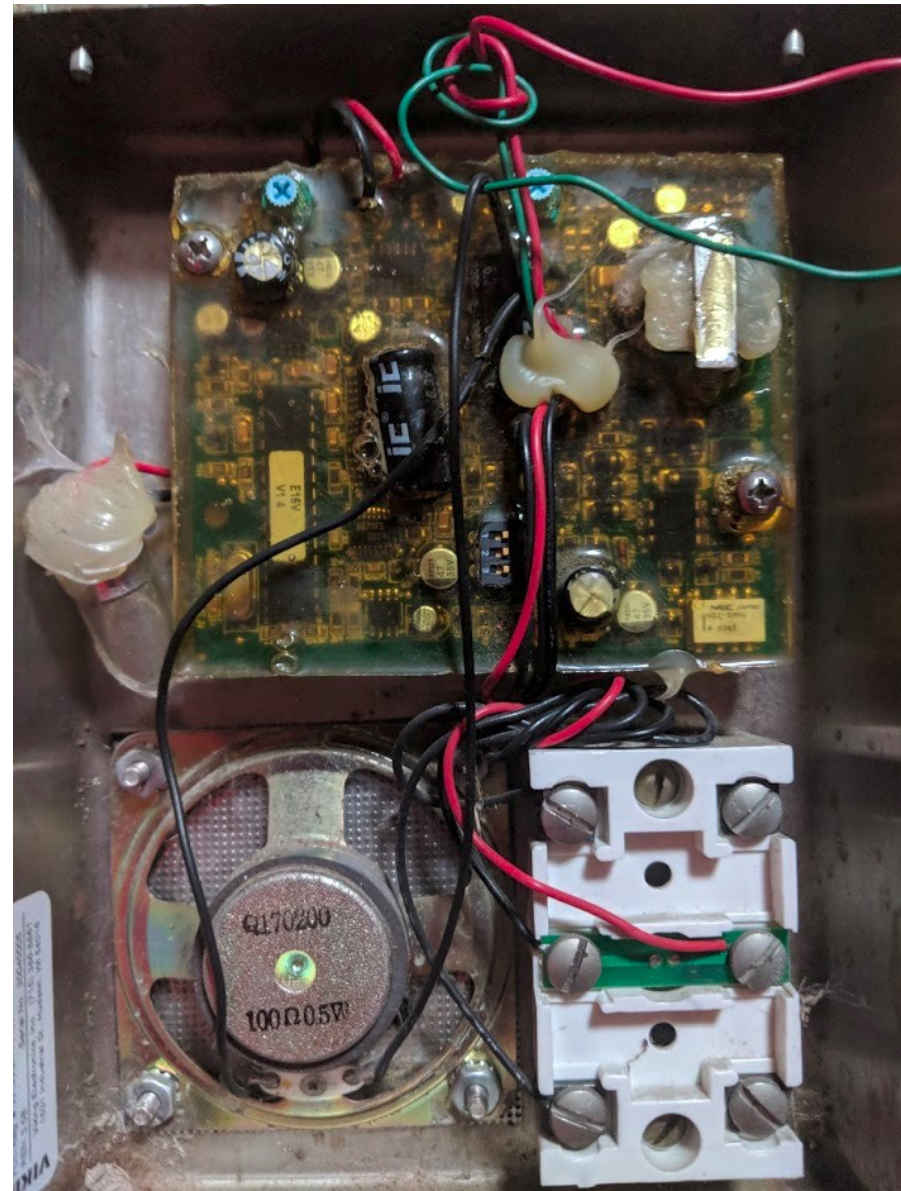
Programming

- **RTFM**
- Use the key pad you
You will need the battery
- Those buttons don't work
the way you think they do.



Programming

- RTFM
- Position 1 connects / disconnects
- Position 3 is programming or “Learn mode”
- Position 2 switch ON-Incoming calls are answered (factory setting)



Programming

Number * **Passcode** * options *#



Default elevator passwords:

- 123456 GAL
- 123456 Lincoln
- 845464 (V-I-K-I-N-G) *some viking products don't even have a password*
- 123456 Viking access password
- 1234 K-Tech
- 35842 T.R.E. Communications
- # 94851 or # 9000000 Janus
- *3*12345678* Master Access Code to change all other codes (Talkaphone)
- *4** guard access (Talkaphone)

HARDER PASSWORDS

- <http://www.datagenetics.com/blog/september32012/index.html>
- 1234 and 123456 are both at the top of the charts for the most common pins.
- The 4 pin code you have a 26.83% chance of getting it right this drops down to 16.12% if 1234 doesn't work.
- A 6 pin code you have a 20.21% chance of getting it right with the top 20 pins though if its not 123456 it drops to 8.53%.

Programming

Number * Passcode * **options** *#



Options?

- Phone number
- Connection time
- Tone or Pulse
- Silence time out
- Dial next number (no answer)
- Dial next number (busy)
- Mute dialing
- Auto talk on ring



Attack!!!!!!

- Denial of service
 - Line no longer functions
 - Phone no longer calls the correct number
- Bypassing line detection

Attack!!!!!!

- Turning the elevator into a covert listening device
- LED problem
- Connection time
- Noise when it picks up

Attack!!!!!!

- You have an open telephone line!
 - Exfiltrate data
 - register the line to a service like google voice
 - Dial some numbers...

900-\$\$\$-\$\$\$\$

- 60 elevators at Brown University
- Each of those elevators has their own telephone line.
- 30 days x 24 hours = 720 hours
- 720 hours x 60 minutes = 43,200 minute in a month
- 43,200 x 60 elevators = 2,592,000 elevator minutes
- **2,592,000 x \$2.55 per minute = \$6,609,600**

Are we all doomed?!?



Look at:

- Monitoring (Why are we having an hour long call?)
- Logging (Why do 30 people call the elevator a day?)
- Alerting (Is there really an emergency?... We should deal with that)

Manufactures

- NO DEFAULT PASSWORDS... Don't allow the most common top 20 pins
- Don't allow remote programming
- Train your call centers for SE attacks

Lets go further!

- Pools
- University campus
- Meeting areas
- THE STAIRS!!!!



Want to learn more?

- Watch TeleChallenge 2018 Walkthrough
- Watch Pit to Penthouse
- RTFM
- C*net
- Bin Rev

If you want to play at home...

- Buy elevator phones on ebay
- Used ones will most likely be broken \$50 - \$100
- New phones cost \$100 - \$300

Want to play around?

- PLA: 914-296-1862
- FUTELL: 503-HOT-1337
- ClownSec Phunhouse: 1-914-495-1365

Thank you

- All of my wonderful infosec friends.
- @plugxor
- @wireghoul
- @SgtHowardPayne
- @deviantollam



Thank you

- **@willcaruana**
- **Call me 617-440-8667 (prank calls only)**
- **hvwill@protonmail.com**